Internship Report

Host Industry: Arch Global Services (Philippines), Inc.
Host Center/Department: Automation Engineering Team, Strategic Analytics Department
Duration: 5 months
Manager/Supervisor: Jerwyn Kyles Mendoza
Manager/Supervisor Contact Details: jmendoza@archgroup.com
Position: Service Delivery Manager

Target Competency	Alignment to Dissertation	Output	Status
Advanced Data Preprocessing & Exploratory Analysis	Address the "Data Spillage" problem highlighted in the dissertation by learning industry standards for strict training/validation separation and data imputation.	Accepted journal article in Digital Chemical Engineering employing industry standards for strict training/validation separation and data imputation. Proof: Accepted Article Production Tracking, SSRN Pre-print, and public GitHub repo	Target Achieved
Production-Grade Software Engineering & MLOps	Improve the computational tractability and reproducibility of the CoBRE optimization model using professional software standards.	In this GitHub repo, we rename CoBRE into ICSOR. It contains the source code for implementing it using professional software standards.	Target Achieved
Advanced Benchmarking & Model Evaluation	Adopt rigorous statistical testing used in NLP/CV (Precision, Recall, F1) to compare the CoBRE model against the ML	Accepted journal article in Digital Chemical Engineering employing rigorous statistical testing to compare the ICSOR (formerly "CoBRE") model against the ML baselines. Proof: Accepted Article	Target Achieved

	baselines (XGBoost, ANN).	Production Tracking, SSRN Pre-print, and public GitHub repo	
Handling Non-Linearity & Optimization Constraints	Gain exposure to solving high-dimensional problems (LLMs/Deep Learning) to better handle the large-scale superstructure optimization in the research.	Accepted journal article in Digital Chemical Engineering employing deep learning (deep multi-layer perceptron) and compared it to the ICSOR (formerly "CoBRE") model. Proof: Accepted Article Production Tracking, SSRN Pre-print, and public GitHub repo	Target Achieved
Technical Communication & Interpretability	Enhance the "Interpretability" argument of the dissertation by learning how to explain "Black-box" AI results.	Accepted journal article in Digital Chemical Engineering introducing ICSOR (formerly "CoBRE") model – a novel interpretable statistical surrogate of the activated sludge process. Proof: Accepted Article Production Tracking, SSRN Pre-print, and public GitHub repo	Target Achieved

Prepared by:

07/31/2026

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